Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation

AI for earth Observation Course - AI4EO - M2 GeoAI

Milestone 4 Project Extension and Additional Results
Geo AI Master 2 – Lebanese University, Faculty of science

June 2026

Final Report

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GENERALIZED FRAMEWORK REPORT

Document Introduction & Scope

This comprehensive final report presents the evaluation of **Task 4** implemented across two diverse applications to demonstrate the versatility, robustness, and adaptability of the generalized pipeline.

The scope of this project encompasses two distinct experimental tracks:

1. **Track 1: Transfer of Agentic XAI from Rice-Yield Regression to Wildfire Classification** — This track evaluates the framework's capability to transition from an agricultural regression problem to a complex, real-world Earth Observation binary classification task utilizing spatial coordinate partitioning.
2. **Track 2: Generalized Agentic XAI Workflow Applied to Appliances Energy Prediction** — This track applies the identical generalized Agentic XAI workflow to a real, dense tabular regression dataset to stress-test the framework's ability to handle continuous, non-synthetic targets under strict temporal sequence constraints.

1-Transfer of Agentic XAI from Rice-Yield Regression to Wildfire Classification

Project: Generalized Agentic XAI workflow applied to Earth-observation wildfire data

Dataset: FiresDataset.parquet

Scope: Framework generality, predictive performance, explanation refinement, and limitations

Team: Task 4 - Team 1

Date: June 13, 2026

Executive Summary

Milestone 4 generalized the paper-inspired workflow so it is no longer locked to rice yield, Radiation, Temperature, or a fixed target name. The notebook successfully transferred the workflow from regression to wildfire classification and generated five evidence stages: a baseline SHAP analysis followed by generalization, subgroup, feature-behavior, and robustness rounds.

Framework conclusion The generalization was technically successful. The notebook is a reusable supervised-tabular Agentic XAI template with configurable data, target, task, domain, identifiers, split strategy, reference predictions, and feature groups.

Model conclusion The new wildfire model failed to achieve reliable spatial generalization. Test balanced accuracy was 0.5837, recall was 0.3891, and 12,882 fire cases were missed. Its value is diagnostic rather than operational.

The retained reference predictions were numerically much stronger, but they are not confirmed as out-of-sample predictions under the same spatial split. Therefore, they cannot yet support a fully fair claim that the earlier model generalizes better.

1. Is the Notebook General?

The notebook is general at the workflow and configuration level. It can load CSV, Excel, or Parquet tabular data; select a target; perform regression or classification; preprocess numerical and categorical features; train a Random Forest; calculate SHAP; generate diagnostics; and create round-specific prompts and recommendation drafts.

Capability	Status	Practical boundary
Dataset and target configuration	Generalized	Path, target, ID, date, and reference columns are configurable.
Task type	Generalized	Regression and classification are supported.
Preprocessing and model	Generalized	Designed for supervised tabular data and tree-based SHAP.
Validation	Configurable	Random, group, or temporal splitting requires suitable columns.
Agentic refinement	Partially automated	Controller selects modules, but LLM_PROVIDER is manual.
Universal any-data support	Not claimed	Images, text, unsupervised tasks, and arbitrary multiclass workflows need extension.

Correct description A generalized, reusable tabular Agentic XAI workflow with a manual language-model checkpoint. It is not yet a fully autonomous agent and is not guaranteed to run unchanged on every possible dataset.

2. Wildfire Dataset and Experimental Design

Setting	Configured value
Dataset	FiresDataset.parquet: 173,991 rows and 51 columns
Target	Fire_Label, binary classification
Domain context	Wildfire occurrence using Earth observation, weather, terrain, and land cover
Eligible features	41 raw model features; 48 transformed features
Split	Spatial/group split using X and Y
Training and test size	134,847 training rows; 39,144 test rows
Model	Random Forest classifier, 300 trees, no maximum depth, random seed 42
SHAP	Approximate TreeSHAP on a reproducible sample of 500 test rows
LLM mode	Manual: prompts and drafts are created, but no API response is generated automatically

The target, coordinate identifiers, date, original fire-label text, and retained reference-result columns were excluded from model training and SHAP. This prevents the framework from directly learning from previous predictions or from the label source.

3. Round 0: Baseline Performance and SHAP

Metric	New generalized model	Retained reference prediction
Accuracy	0.5687	0.9523
Balanced accuracy	0.5837	0.9510
Precision	0.6722	0.9454
Recall	0.3891	0.9672
F1	0.4929	0.9562
ROC-AUC	0.6892	0.9834

Prediction result	New generalized model	Reference prediction
True negatives	14,054	16,879
False positives	4,002	1,177
False negatives	12,882	691
True positives	8,206	20,397

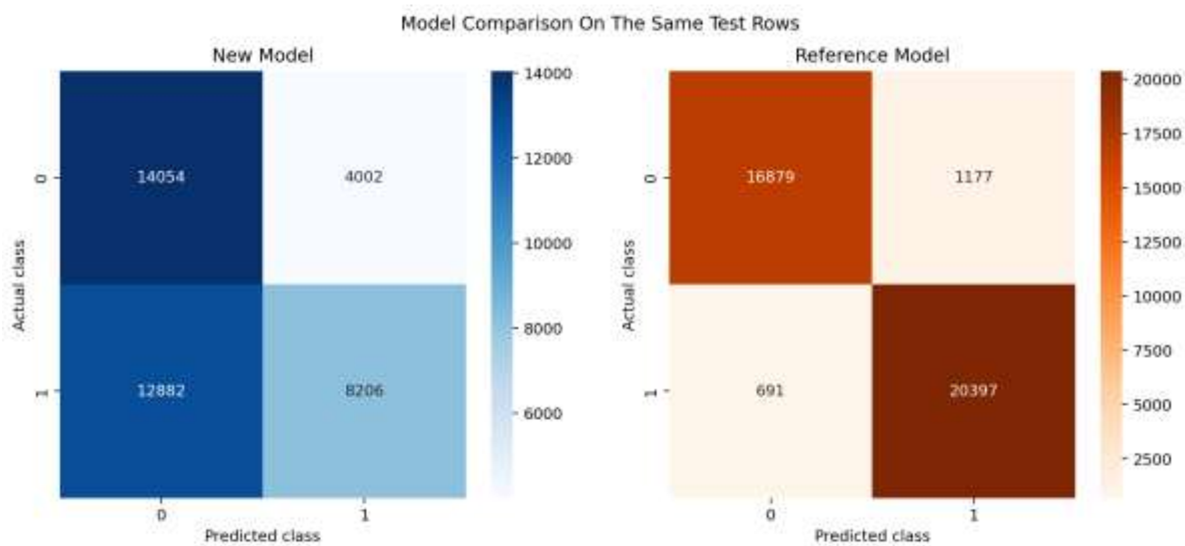

Figure 1. Prediction agreement and disagreement between the generalized model and retained reference outputs.

Across the 39,144 test rows, both systems were correct for 21,283 records. The reference alone was correct for 15,993 records, the new model alone for 977, and both were wrong for 891.

Fairness warning REFERENCE_IS_OUT_OF_SAMPLE is set to False. The reference predictions may have been generated under a different or less restrictive evaluation design. The comparison is numerically informative but cannot yet establish a fair generalization advantage.

The strongest SHAP features were Elevation__Low, Elevation, Aspect, LUC_majori, and Slope. This domain-specific ranking confirms that the generalized workflow is not reusing rice-specific fields.

4. Refinement-Round Findings

Round 1: Generalization and Calibration

Training balanced accuracy was 0.9992, while test balanced accuracy was 0.5837. The gap of 0.4155 is evidence of severe overfitting. Bootstrap analysis estimated test balanced accuracy at 0.5834 with a 95% interval from 0.5788 to 0.5874. The Brier score was 0.2371 and expected calibration error was 0.1459, indicating that predicted probabilities were not well calibrated.

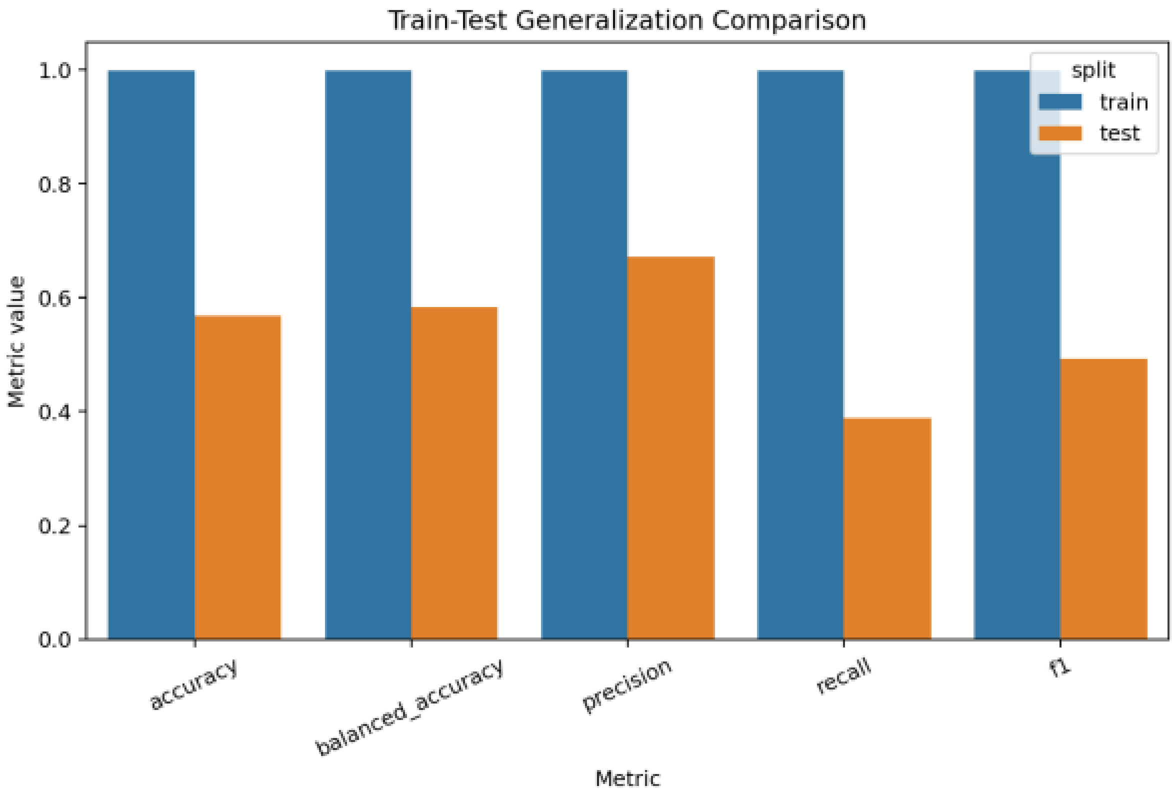

Figure 2. Train-test performance comparison used to diagnose model generalization.

Round 2: Error, Subgroup, and Spatial Audit

The worst observed subgroup was Elevation between 900.924 and 1114.59, containing 9,778 records. Its error rate was 0.6153, balanced accuracy was 0.5131, and fire recall was approximately 0.0997. This shows that global metrics conceal severe location- and terrain-dependent failure.

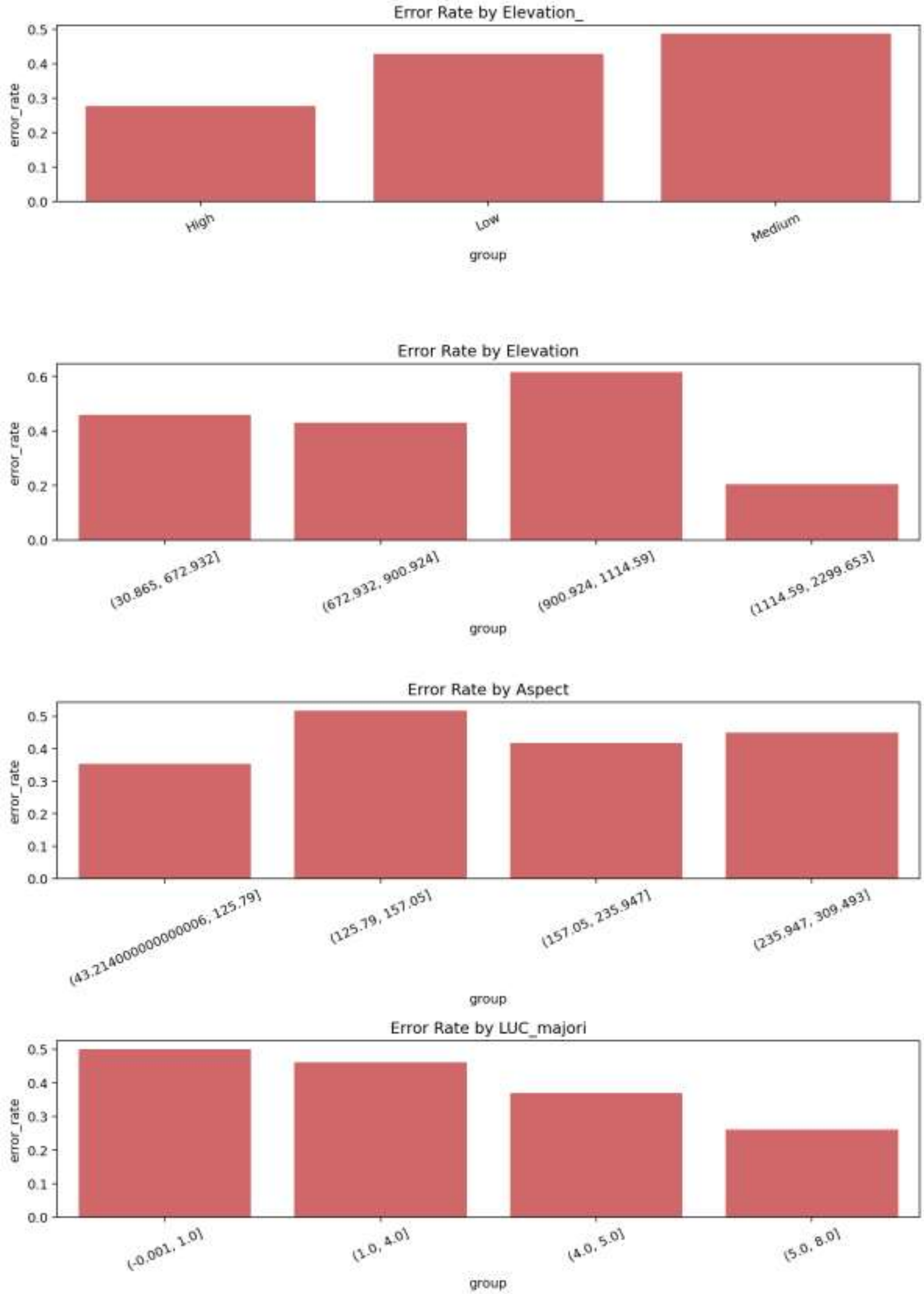

Figure 3. Subgroup performance audit across influential terrain and land-cover variables.

Round 3: Feature Behavior and Co-Movement

Elevation__Low was the strongest analyzed feature, with mean absolute SHAP = 0.0613 and a value-SHAP Spearman correlation of 0.8560. Elevation__Low and numeric Elevation had the strongest SHAP co-movement, 0.4260. This suggests that categorical elevation bands and numeric elevation represent overlapping information and can divide or duplicate apparent importance.

Interpretation boundary SHAP co-movement is a screening indicator. It must not be reported as a causal effect or a formal interaction estimate.

Round 4: Robustness and Sensitivity

On 2,000 held-out rows, the baseline balanced accuracy was 0.5837. Permuting Elevation__Low reduced the score by 0.0401, and permuting numeric Elevation reduced it by 0.0315. In contrast, permuting Aspect increased balanced accuracy by approximately 0.0327. Aspect therefore appears unstable or harmful under the spatial holdout.

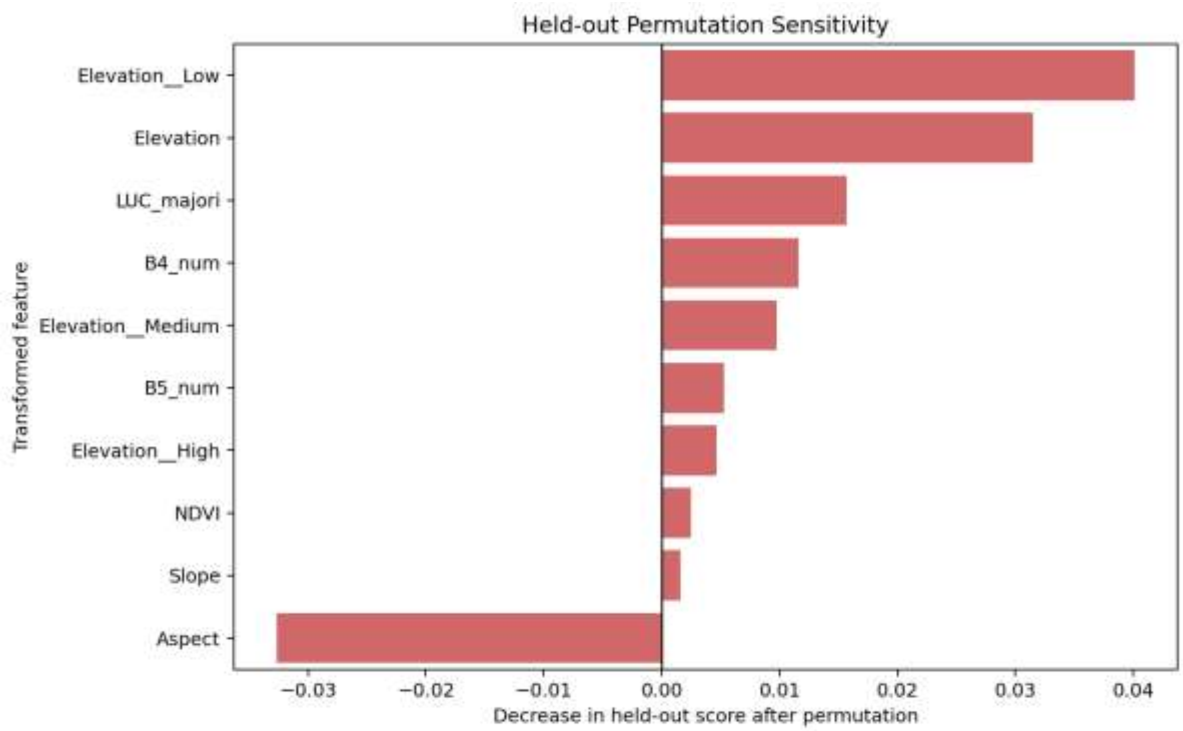

Figure 4. Held-out permutation sensitivity for the leading transformed features.

5. Why the New Wildfire Model Failed

Severe overfitting Near-perfect training metrics did not transfer to spatially separated test data.

Low fire detection Recall of 0.3891 means the model identified fewer than four of every ten positive fire records in the test set.

Subgroup instability Some elevation groups had fire recall near zero, showing that the model's behavior changes sharply across terrain conditions.

Feature redundancy Numeric and banded elevation variables describe the same underlying concept and may encourage brittle splits.

Aspect representation Raw Aspect is circular, where values near 0 and 360 degrees are close in reality but distant numerically. Its negative permutation result indicates that the current representation may damage generalization.

Operational conclusion The generalized model is not suitable for wildfire decision support in its current form. The failure is a model-performance finding, not a failure of the generalized XAI workflow.

6. Agentic XAI Status

The notebook implements a structured refinement controller. Each round reads the preceding evidence, identifies an analytical gap, selects a general analysis module, generates new figures and tables, and writes a prompt plus recommendation draft for the next review.

Agentic component	Current status
Evidence-driven module selection	Implemented for Rounds 1-4
Round-specific figures and tables	Implemented
Prompt generation	Implemented
Recommendation draft generation	Implemented
Automatic LLM self-reflection	Not active; LLM_PROVIDER is manual
Closed-loop autonomous refinement	Not yet complete

The correct claim is therefore that Milestone 4 implements a generalized human-in-the-loop Agentic XAI framework. To claim full autonomous agency, the same configured LLM must generate and reflect on each recommendation, with its output passed into the next round.

7. Relationship to Task 3 and the Original Paper

Task 3 showed that a synthetic rice dataset could reproduce a similar SHAP story while failing to reproduce the paper's validation performance. It also showed that additional refinement rounds can reduce groundedness and that duplicate records can create misleadingly high validation scores.

Task 4 extends these lessons into a different domain and task type. The workflow moved from rice-yield regression to wildfire classification, generated different feature explanations, and selected new analytical modules. This is strong evidence that the framework is no longer tied to the rice case study.

Combined scientific result The project contributes more through disciplined iterative diagnosis than through improved prediction. Across both milestones, the framework exposed overfitting, unsupported recommendation expansion, duplicate leakage, subgroup failure, calibration weakness, feature redundancy, and instability.

8. Recommended Next Validation

Fair reference comparison Recreate the earlier wildfire Random Forest using the same spatial train-test split, identical test rows, feature exclusions, and positive-class definition.

Terrain-feature ablation Compare numeric Elevation alone, categorical Elevation_ alone, and both together.

Circular Aspect encoding Replace raw Aspect with sine and cosine components and compare against removing Aspect completely.

Model regularization Tune maximum depth, minimum samples per leaf, maximum features, and class weighting using grouped spatial validation.

Probability calibration Evaluate calibration on untouched spatial folds before interpreting fire probabilities as risk estimates.

Complete the LLM loop Use one fixed language model, prompt configuration, temperature, and evaluation rubric for all rounds, then repeat the complete sequence and score outputs with multiple reviewers.

9. Limitations

Reference provenance The retained reference predictions are not confirmed as out-of-sample under the current spatial split.

Single model family The current generalized run evaluates one Random Forest configuration rather than a tuned model-selection process.

Approximate SHAP sample Explanations were computed on 500 test rows for practical local execution, not the full test set.

Manual language-model mode The generated recommendation files contain evidence-grounded drafts and placeholders rather than completed automatic LLM reflection.

Binary classification focus The wildfire implementation does not by itself establish complete support for arbitrary multiclass, image, text, or unsupervised datasets.

10. Recommended Final Statement

Suggested presentation wording We generalized the paper-inspired Agentic XAI workflow from rice-yield regression to a configurable tabular framework and applied it to wildfire classification. The transfer was successful at the workflow level: the system generated SHAP explanations, selected evidence-driven refinement modules, created new diagnostics, and produced round-specific prompts and recommendation drafts. The new wildfire model itself showed severe spatial overfitting and low fire recall, so it should not be presented as superior to the previous prediction system. Its main value is that the generalized Agentic XAI process revealed where and why the model fails, while also identifying the experiments needed for a fair next comparison.

2-Generalized Agentic XAI Workflow Applied to Appliances Energy Prediction

Project: Generalized Agentic XAI workflow applied to real tabular regression data

Dataset: energydata_complete.csv

Scope: Framework generality, regression performance, split sensitivity, SHAP explanation, and limitations

Executive Summary

Milestone 4 was extended from wildfire classification to a real regression dataset: Appliances Energy Prediction. This experiment tests whether the generalized Agentic XAI workflow can handle a non-rice, non-synthetic, continuous target while still producing SHAP explanations, refinement rounds, figures, prompts, recommendation drafts, and evidence files.

Framework conclusion The transfer was technically successful. The notebook remains configurable for supervised tabular regression or classification and is not locked to Yield, Radiation, Temperature, rice fields, wildfire labels, or fixed column names.

Model conclusion The random-row regression score was moderate, but the refinement workflow revealed severe validation sensitivity. The model should not be presented as a reliable future forecasting system.

Scientific value The experiment is useful because it shows how Agentic XAI can uncover an overly optimistic baseline and redirect the analysis toward temporal generalization, subgroup errors, and robustness.

1. Is the Notebook General?

The notebook is general at the workflow and configuration level. It can load CSV, Excel, or Parquet data, select a target column, switch between regression and classification, exclude ID/date/reference fields, generate date-derived features, preprocess numerical and categorical variables, train a Random Forest model, compute SHAP, and generate round-specific figures, evidence, prompts, and recommendation drafts.

Capability	Status	Practical boundary
Dataset path and target	Generalized	Changed through DATASET_PATH and TARGET_COLUMN.
Task type	Generalized	Regression and classification are supported.
Date handling	Generalized	Date columns can be excluded directly and converted into reusable calendar features.
Validation	Configurable	Random, grouped, or temporal splitting must match the deployment question.
Recommendations	Human-in-the-loop	The code writes evidence drafts; final LLM text is manual unless an API adapter is added.

2. Dataset and Experimental Design

Setting	Configured value
Dataset	energydata_complete.csv: 19,735 rows and 29 original columns
Target	Appliances, continuous regression target
Domain context	Appliance energy consumption using indoor climate, outdoor weather, lighting, and time context
Excluded fields	date as raw timestamp; rv1 and rv2 because they are duplicate random variables
Generated features	Cyclical time-of-day, week, month, and weekend indicators
Primary split	Random 80/20 baseline, challenged by grouped-date and future-period stress tests
Model	RandomForestRegressor with 300 trees
SHAP sample	1,000 held-out rows for practical local execution

This dataset is closer to the original rice case mathematically because it is a tabular regression problem. It is not Earth observation data; the wildfire classification experiment remains the stronger EO-domain case.

3. Round 0: Baseline Regression and SHAP

Metric	Train	Random-row test
R2	0.9461	0.6171
RMSE	23.9501	61.9039
MAE	11.2400	29.1678

The random-row test R2 was 0.6171, with RMSE 61.90 and MAE 29.17. This is a moderate conventional regression result, but it is not enough to claim deployable future forecasting performance.

The strongest SHAP features were time-of-day sine, time-of-day cosine, lights, weekly timing, pressure, room temperature T3, and room humidity RH_3. The model therefore learned behavior and schedule patterns in addition to environmental relationships.

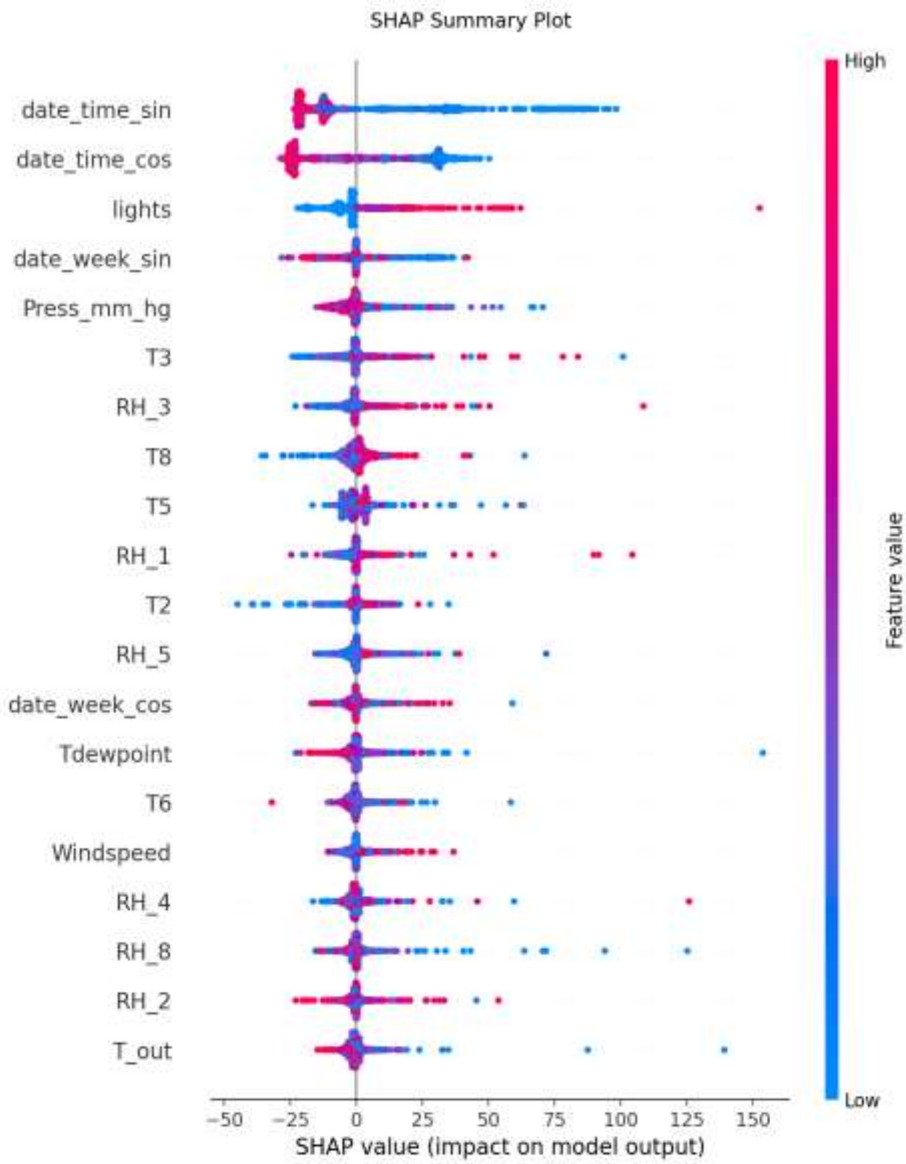

Figure 1. Round 0 SHAP beeswarm showing that time context, lighting, pressure, temperature, and humidity drive the random-row baseline.

4. Refinement-Round Findings

Round 1: Split Sensitivity

Evaluation design	Test R2	RMSE	MAE	Interpretation
random	0.6171	61.90	29.17	Moderate
grouped_dates	-0.0527	107.00	63.52	Failure
temporal_future	-3.6599	196.52	159.47	Failure

This is the most important result. The same workflow that first produced a moderate random-row result then exposed that the model does not generalize well to separated dates or a future time period.

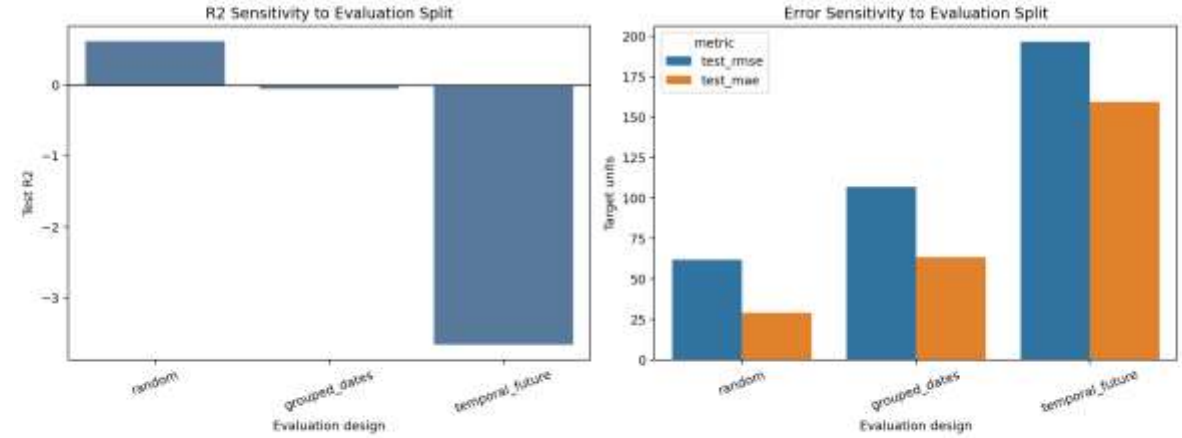

Figure 2. Round 1 validation sensitivity: random-row validation is optimistic, while grouped-date and future-period tests fail.

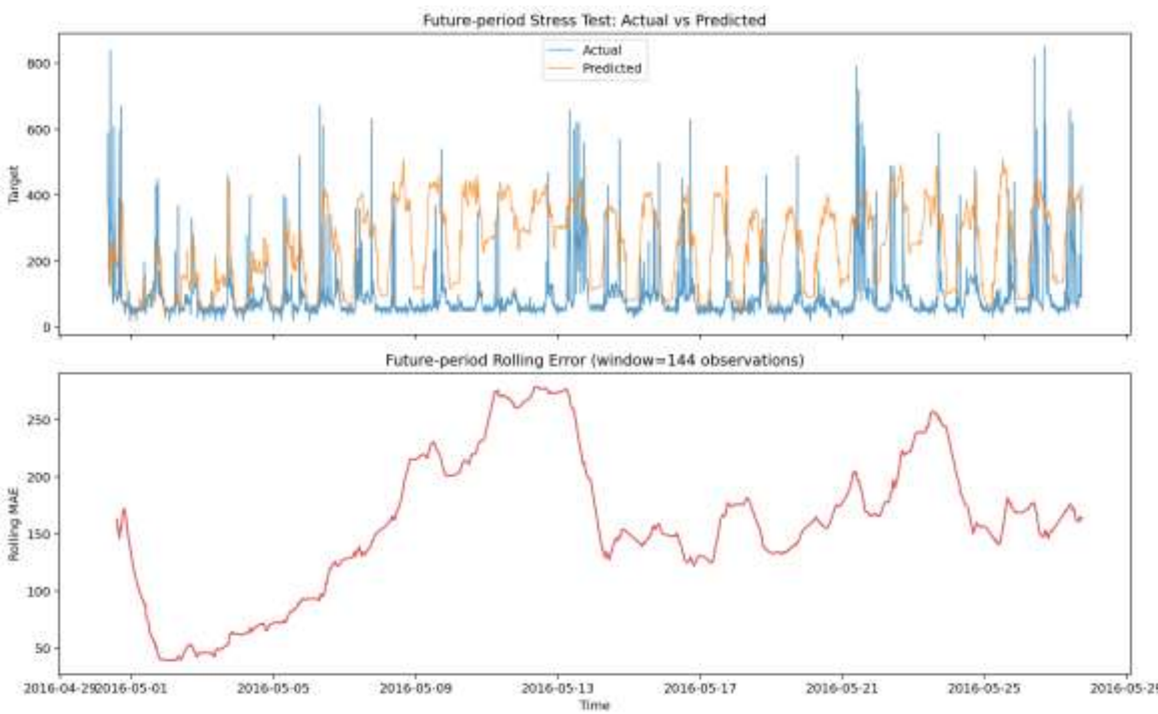

Figure 3. Future-period stress test showing overprediction and large rolling error across the held-out future period.

Round 2: Subgroup Error Audit

The worst audited subgroup was lights = 30 with 104 observations and MAE = 50.43. Higher lighting levels produced larger errors than the lights = 0 group, suggesting that the model struggles during more active household periods.

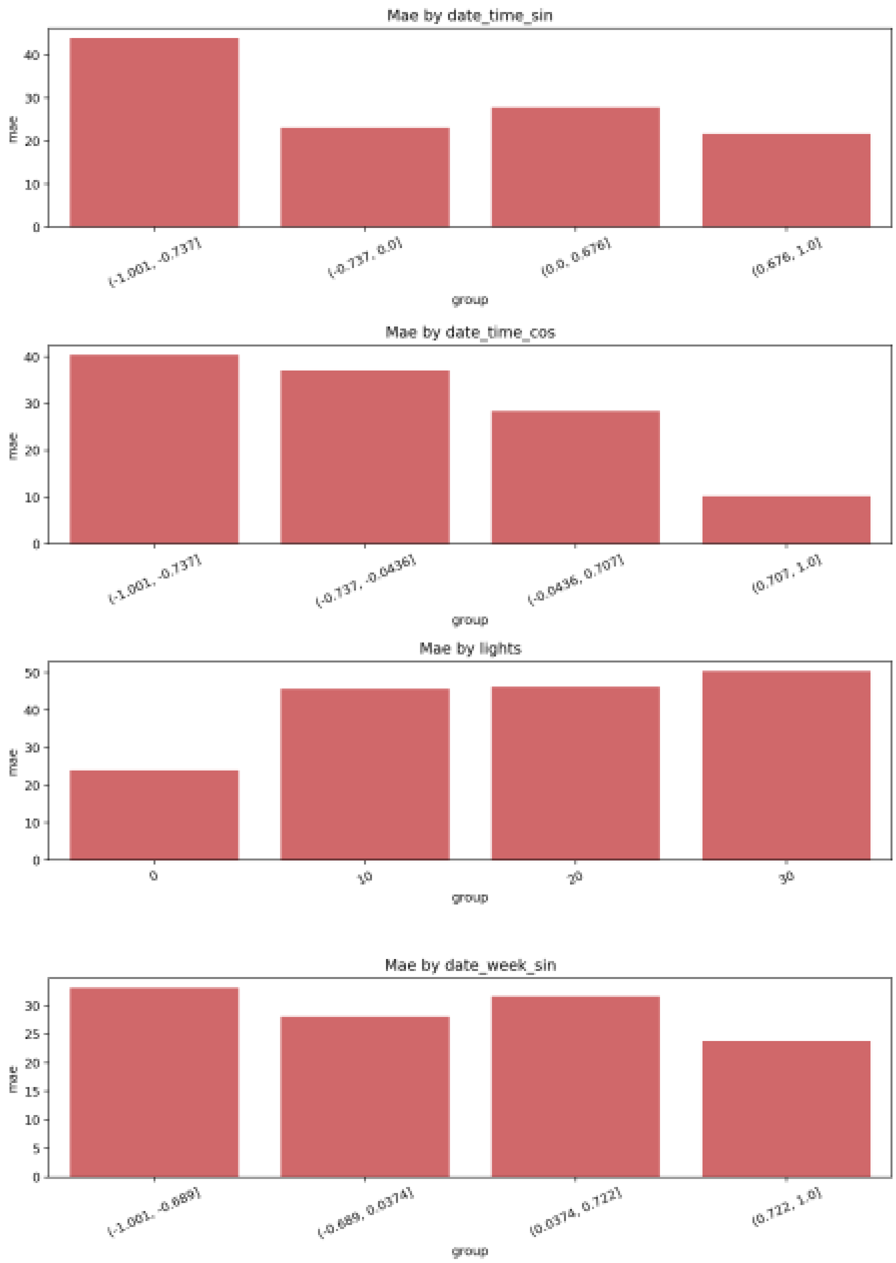

Figure 4. Round 2 subgroup audit showing where regression errors concentrate.

Round 3: Feature Behavior and Co-Movement

The strongest analyzed feature effect was date_time_sin with mean absolute SHAP = 24.1703 and value-SHAP Spearman correlation = -0.8056. The strongest contribution co-movement pair was date_time_sin and date_time_cos with Spearman co-movement = 0.4315.

Interpretation boundary: SHAP co-movement is only a screening indicator. It must not be reported as a causal effect or a formal interaction estimate.

Round 4: Robustness and Explanation Stability

On 2,000 held-out rows, the baseline R2 was 0.6059. Permuting date_time_cos reduced R2 by 0.4394. The least stable reported top feature was T6, with top-10 frequency 0.0000.

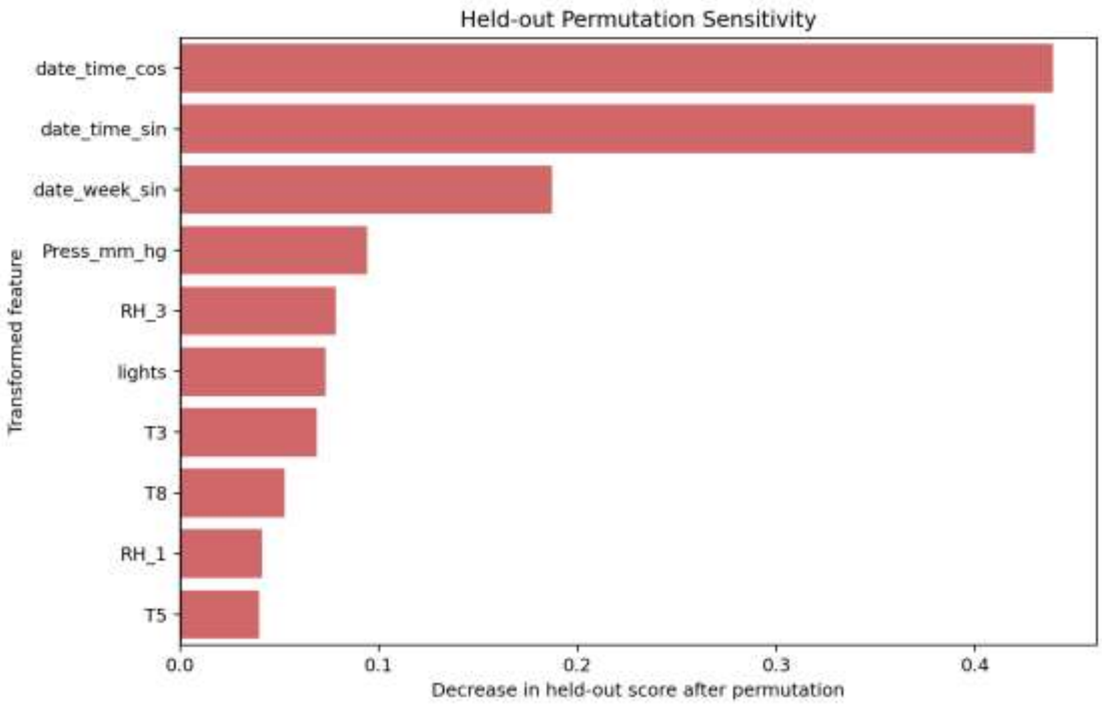

Figure 5. Round 4 permutation sensitivity confirms strong dependence on time-derived features.

5. Why the Regression Model Failed

Temporal leakage risk Random rows place neighboring ten-minute observations into both training and testing, making performance look stronger than it is.

Future-period failure The future-period R2 was -3.6599, meaning the model was worse than predicting the average target.

Systematic overprediction In the future-period test, actual mean Appliances was 96.38 while predicted mean was 241.71.

Behavioral dependence The leading SHAP features are time and lighting, indicating strong dependence on household schedule/activity patterns.

Operational conclusion The model is not suitable for future energy forecasting in its current form. The workflow is valuable because it detected this failure.

6. Agentic XAI Status

Agentic component	Current status
Evidence-driven module selection	Implemented for Rounds 1-4
Round-specific figures and tables	Implemented
Prompt generation	Implemented with three reusable prompt templates

Recommendation draft generation	Implemented as Markdown evidence drafts
Automatic LLM reflection	Not implemented; LLM_PROVIDER is manual
Full autonomous agency	Not claimed

The correct claim is that Milestone 4 implements a generalized human-in-the-loop Agentic XAI framework. To claim full autonomous agency, the same configured LLM must generate and reflect on each recommendation, with its response passed into the next round under fixed prompt and temperature settings.

7. Relationship to Wildfire, Task 3, and the Original Paper

The appliances experiment is stronger than synthetic agriculture as a real-data regression test, but weaker than wildfire in Earth-observation relevance. Together, the two Task 4 experiments show two complementary things: wildfire tests EO-domain transfer and classification, while appliances tests real tabular regression transfer.

The result supports the project interpretation that the paper's important contribution is not the rice model itself, but the iterative refinement process: SHAP explanation, reflection, analytical gap identification, new figures, and updated recommendations.

8. Recommended Next Validation

- Use rolling or expanding-window time-series validation before making forecasting claims.
- Compare the Random Forest against a simple time-of-day/week baseline.
- Evaluate models with and without generated calendar features to separate schedule learning from environmental effects.
- Add occupancy or appliance-operation variables if the objective is operational forecasting.
- Consider log-transforming the target or separately modeling high-consumption events.
- Use one fixed LLM, prompt configuration, temperature, and evaluation rubric for all recommendation rounds.

9. Limitations

Non-EO domain Appliances energy prediction is environmental/building-energy data, not Earth observation.

Single building and period The dataset covers one monitored setting over approximately 4.5 months.

Manual LLM mode Recommendation files are evidence-grounded drafts, not completed automatic LLM outputs.

One model family The run evaluates one Random Forest configuration rather than a full model-selection study.

Approximate SHAP sample SHAP was computed on 1,000 held-out rows for local practicality.

10. Recommended Final Statement

Suggested presentation wording We generalized the paper-inspired Agentic XAI workflow to a configurable tabular framework and applied it to Appliances Energy Prediction as a real regression case. The transfer was successful at the workflow level: the system generated SHAP explanations, selected evidence-driven refinement modules, created new diagnostics, and produced round-specific prompts and recommendation drafts. The model achieved a moderate random-row R2 of 0.617, but failed under grouped-date and future-period validation. Therefore, the result should not be presented as a strong forecasting model; its main value is that the Agentic XAI process revealed where the baseline explanation was over-optimistic and what validation is needed next.